

Expressing Concentration

Mass Percent

$$\text{Percent by mass} = \frac{\text{mass component}}{\text{mass solution}} \times 100\%$$

↑ solvent
↑ solute

40% $\text{HNO}_3(\text{aq})$ by mass

↓
solution = 100 g

40 g HNO_3

60 g H_2O

$$\text{mass}_1 \% + \text{mass}_2 \% + \dots = 100\%$$

2 Parts per million. ppm

$$\text{ppm} = \frac{\text{mass component}}{\text{mass solution}} \times 10^6$$

$$1 \text{ ppm} = 1 \text{ } \mu\text{g/g} = 1 \text{ mg/kg} = 1 \text{ mg/L} = 1 \text{ } \mu\text{g/mL}$$

5 ppm $\text{Na}^+(\text{aq}) \rightarrow$ 1000 000 g solution

5g Na^+

1000 095 g
Solvent
& H_2O

3 part per Billion, ppb

$$\text{ppb} = \frac{\text{mass component}}{\text{mass solution}} \times 10^9$$

$$1 \text{ ppb} = 1 \mu\text{g/kg} = 1 \mu\text{g/L} = 1 \text{ ng/mL} = 1 \text{ ng/g}$$

10 ppb $\text{Ca}^{+2}_{(\text{aq})}$

4 Mole Fraction (χ)

$$\chi = \frac{\text{mol component}}{\text{total mol}} = \frac{n_i}{n_t}$$

$$\chi_{\text{solute}} + \chi_{\text{solvent}} = 1$$

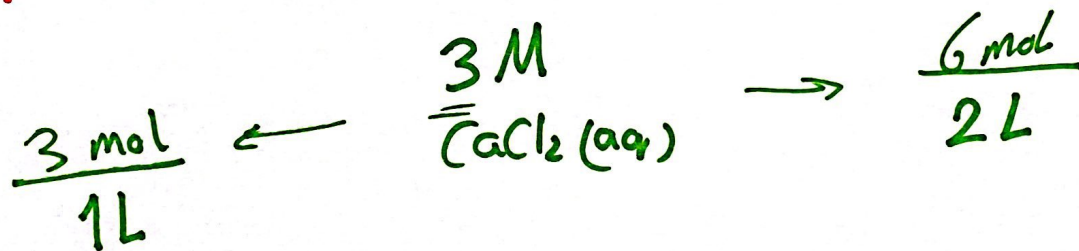
$$\chi_{\text{solute}} = 0.5$$

$$\chi_{\text{solvent}} = 1 - 0.5 = \underline{0.5}$$

5 Molarity (M)

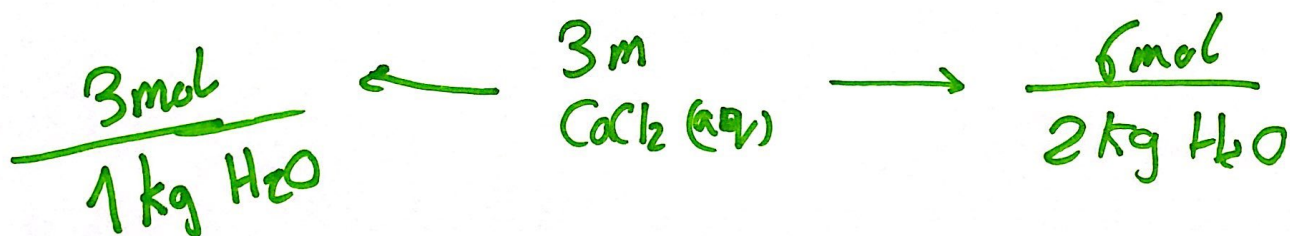
$$M = \frac{\text{moles solute}}{\text{Volume solution (L)}} = \frac{n}{V(L)} \quad \begin{matrix} [\text{mol/L}] M \\ [n \text{ mol/L}] mM \end{matrix}$$

- temp dependent.



6 Molarity (m)

$$m = \frac{\text{moles solute}}{\text{Kg solvent}} = \frac{n}{m(\text{kg})} \quad \frac{\text{mol/kg}}{\boxed{m}}$$



2.5g sample of groundwater was found to contain 5.4 μ g of Zn^{+2} . What is the concentration of Zn^{+2} in ppm.

$$\text{ppm} = \frac{\text{mass component}}{\text{mass solution}} \times 10^6$$

$$= \frac{5.4 \times 10^{-6}}{2.5} \times 10^6$$

$$= \boxed{2.16 \text{ ppm}}$$

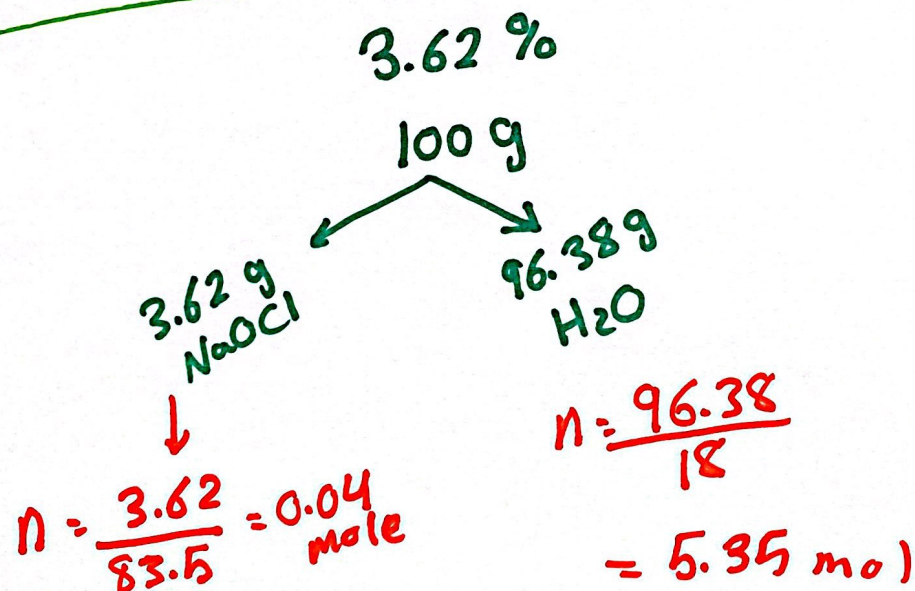
What is the molality of a solution made by dissolving 36.5 g of naphthalene [$C_{10}H_8$, $M = 128 \text{ g/mol}$] in 425 g of toluene C_7H_8 ?

$$m = \frac{\text{moles of solute}}{\text{kg (solvent)}} = \frac{0.29}{425 \times 10^{-3}} = 0.68 \frac{\text{mol}}{\text{kg}}$$

$$\text{moles} = \frac{\text{mass}}{M_m} = \frac{36.5}{128} = 0.29 \text{ mol}$$

A commercial bleach solution contains 3.62 mass % NaOCl in water. Calculate the mole Fraction of NaOCl ($M = 83.5 \text{ g/mol}$)

$$X_{\text{NaOCl}} = \frac{\text{moles NaOCl}}{\text{total moles}} = \frac{0.04}{0.04 + 5.35} = 7.4 \times 10^{-3}$$



A commercial bleach solution contains 3.62 mass % NaOCl in water. Calculate the molality of NaOCl ($M = 83.5 \text{ g/mol}$)?

$$m = \frac{\text{moles of solute}}{\text{kg solvent}} = \frac{0.04}{96.38 \times 10^{-3}} = 0.42 \text{ mol/kg}$$

The concentration of $\text{Pb}(\text{NO}_3)_2$ in 0.626 M solution is —
molal. The density of the solution is 1.202 g/mL.

$$m = \frac{\text{moles of solute}}{\text{kg solvent}} = \frac{0.626 \text{ mol}}{0.9947 \text{ kg}} = 0.63 \text{ mol/kg (m)}$$

$$M = \frac{\text{moles of solute}}{V_{\text{solution (L)}}$$

$$0.626 \text{ M} = \frac{0.626 \text{ mol}}{1 \text{ L}}$$

$$d = \frac{\text{mass}}{\text{volume}}$$

$$1.202 = \frac{m}{1000} \Rightarrow m = \underline{\underline{1202 \text{ g}}}$$

$$n = \frac{\text{mass}}{\text{MM}}$$

$$\text{mass solution} = \text{mass solvent} + \text{mass solute}$$

$$1202 = X + (0.626 \times 331.2)$$

$$1202 - 207.3312$$

$$X = 994.7 \text{ g}$$

$$= \underline{\underline{0.9947 \text{ kg}}}$$

A Sulfuric acid solution containing 571.6 g of H_2SO_4 per liter of Solution has a density of 1.329 g/ml. Calculate:

□ the mass percent.

$$\text{mass \%} = \frac{\text{mass } \text{H}_2\text{SO}_4}{\text{mass solution}} \times 100\%$$

$$= \frac{571.6 \text{ g}}{1329 \text{ g}} \times 100\%$$

$$= \underline{43.01\%}$$

$$d = \frac{\text{mass}}{V}$$

$$1.329 = \frac{m}{1000}$$

$$\text{mass} = \underline{\underline{1329 \text{ g}}}$$

b) the mole fraction.

$$X = \frac{\text{moles H}_2\text{SO}_4}{\text{moles total}} = \frac{5.833}{5.833 + 42.03} = \boxed{0.1219}$$

$$- n_{\text{H}_2\text{SO}_4} = \frac{571.6}{98} = 5.833 \text{ mol}$$

$$\begin{aligned} - \text{mass H}_2\text{O} &= \text{mass solution} - \text{mass H}_2\text{SO}_4 \\ &= 1329 - 571.6 = 757.4 \text{ g} \end{aligned}$$

$$- n_{\text{H}_2\text{O}} = \frac{757.4}{18} = 42.03 \text{ mol}$$

c) the molality.

$$m = \frac{\text{moles H}_2\text{SO}_4}{\text{kg H}_2\text{O}} = \frac{5.833}{0.7574} = \underline{7.701\text{ m}}$$

d) the molarity of H_2SO_4 in this solution

$$M = \frac{n \text{ H}_2\text{SO}_4}{V(\text{L})} = \frac{5.833}{1} = \underline{\underline{5.833\text{ M}}}$$

A solution contains Cu^{+2} (63.55 g/mol) ions at a concentration of $3 \times 10^{-4} \text{ M}$. What is the Cu^{+2} concentration in ppm?

$$V = 1 \text{ L}$$

$$\frac{\text{mol}}{\text{L}}$$



$$\frac{\text{mg}}{\text{L}}$$

$$n = \frac{\text{mass}}{\text{molar mass}}$$

$$3 \times 10^{-4} = \frac{\text{mass}}{63.55} \Rightarrow \text{mass} = 0.019 \text{ g} \\ = 19.068 \text{ mg}$$

$$19.1 \text{ ppm}$$